

A General Theory of Strength for Anisotropic Materials

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An operationally simple strength criterion for anisotropic materials is developed from a scalar function of two strength tensors. Differing from existing quadratic approximations of failure surfaces, the present theory satisfies the invariant requirements of coordinate transformation, treats interaction terms as independent components, takes into account the difference in strengths due to positive and negative stresses, and can be specialized to account for different material symmetries, multi-dimensional space, and multi-axial stresses. The measured off-axis uniaxial and pure shear data are shown to be in good agreement with the predicted values based on the present theory.

INTRODUCTION

FOR THE PURPOSE of materials characterization and design, an operationally simple strength criterion for filamentary composites is essential. Strength is an elusive and ambiguous term. It covers many aspects associated with the failures of materials such as fracture, fatigue and creep, under quasi-static or dynamic loading, exposed to inert or corrosive environments, subjected to uni- or multi-axial stresses, in 2 or 3-dimensional geometric configurations, etc. Failures of composites are further complicated by a multitude of independent and interacting mechanisms which include filament breaks and micro-buckling, delamination, dewetting, matrix cavitation and crack propagation. An operationally simple strength criterion cannot possibly explain the actual mechanisms of failures. It is intended only as a useful tool for materials characterization, which determines how many independent strength components exist and how they are measured; and for design which

requires a relatively simple method of estimating the load-carrying capacity of a structure.

There have been numerous strength criteria in existence and additional ones are frequently being proposed. In the ASTM's *Composite Materials: Testing and Design* [1], various strength criteria are used or alluded to by nearly one half of all the contributors. Nearly all of them agree with one another with reference to the principal strengths; i.e., those uniaxial and pure shear test data measured along material symmetry axes. Those strengths are the intercepts of the failure surface with the coordinate axes in the stress-space. The disagreements among existing criteria usually occur in the combined-stress state; i.e., in the space away from the coordinate axes of the failure surface. Since reliable experimental data in the combined-stress state are emerging rapidly, it is, therefore, timely to examine the validity and utility of existing strength criteria, and to propose a general theory.

BASIC ASSUMPTION

The basic assumption of our strength criterion is that there exists a failure surface in the stress-space in the following scalar form:

$$f(\sigma_k) = F_i \sigma_i + F_{ij} \sigma_i \sigma_j = 1 \quad (1a)$$

where the contracted notation is used; and $i, j, k = 1, 2, \dots, 6$; F_i and F_{ij} are strength tensors of the second and fourth rank, respectively. Equation (1a) in expanded or long-hand form is:

$$\begin{aligned} & F_1 \sigma_1 + F_2 \sigma_2 + F_3 \sigma_3 + F_4 \sigma_4 + F_5 \sigma_5 + F_6 \sigma_6 \\ & + F_{11} \sigma_1^2 + 2F_{12} \sigma_1 \sigma_2 + 2F_{13} \sigma_1 \sigma_3 + 2F_{14} \sigma_1 \sigma_4 + 2F_{15} \sigma_1 \sigma_5 + 2F_{16} \sigma_1 \sigma_6 \\ & + F_{22} \sigma_2^2 + 2F_{23} \sigma_2 \sigma_3 + 2F_{24} \sigma_2 \sigma_4 + 2F_{25} \sigma_2 \sigma_5 + 2F_{26} \sigma_2 \sigma_6 \\ & + F_{33} \sigma_3^2 + 2F_{34} \sigma_3 \sigma_4 + 2F_{35} \sigma_3 \sigma_5 + 2F_{36} \sigma_3 \sigma_6 \\ & + F_{44} \sigma_4^2 + 2F_{45} \sigma_4 \sigma_5 + 2F_{46} \sigma_4 \sigma_6 \\ & + F_{55} \sigma_5^2 + 2F_{56} \sigma_5 \sigma_6 \\ & + F_{66} \sigma_6^2 = 1 \end{aligned} \quad (1b)$$

The linear term in σ_i takes into account internal stresses which can describe the difference between positive- and negative-stress induced failures. The quadratic terms $\sigma_i \sigma_j$ define an ellipsoid in the stress-space. In our basic assumption in Equation (1), we ignored higher order terms, e.g., term $F_{ijk} \sigma_i \sigma_j \sigma_k$ in the strength criterion is not practical from the operational standpoint because the number of components in a 6th-rank tensor run into the hundreds. In addition, having cubic terms, the failure surface becomes open-ended.

Several features of our proposed strength criterion are as follows:

(1) It is a scalar equation and is automatically invariant. Interactions among all stress components are independent material properties. In criteria by quadratic approximations such as Hill's [2], interactions are fixed (not independent). In the maximum stress or maximum strain criterion, six simultaneously equations are required and interactions are not admissible.

(2) Since strength components are expressed in tensors, their transformation relations and the associated invariants are well established. In particular, the transformation relations in terms of multiple angles, similar to those developed for the elastic stiffness [3], are useful tools for the understanding of strength tensors.

(3) The symmetry properties of the strength tensors and the number of independent and non-zero components can be rigorously described similar to other well-established properties of anisotropic materials, such as the elastic compliance matrix. The number of spacial dimensions and multiaxial stresses are determined by selecting the proper range of the indices among 1 to 6. General anisotropy and 3-dimensional space present no conceptual difficulty.

(4) Knowing the transformation relations we can readily rotate the material axes from F_i to F'_i and F_{ij} to F'_{ij} in Equation (1), or equivalently rotate in the opposite direction to change the applied stresses from σ_i to σ'_i when we want to study the off-axis or transformed properties. Most existing criteria are limited to specially orthotropic materials. These criteria can only be applied by transforming the external stresses to the material axes. Rotation of the material axes cannot be done because the transformations of the strength criteria are not known.

(5) Being invariant, Equation (1) is valid for all coordinate systems when it is valid for one coordinate system. Such validity holds for curvilinear coordinates as well with only minor adjustments.

(6) Certain stability conditions are incorporated in the strength tensors. The magnitude of interaction terms are constrained by the following inequality:

$$F_{ii}F_{jj} - F_{ij}^2 \geq 0 \quad (2)$$

where repeated indices are NOT summations for this equation; and $i, j = 1, \dots, 6$. F_{ii} is simply one of the diagonal terms. To be physically meaningful, all diagonal terms must be positive; the off-diagonal of interaction terms may be positive or negative depending on the nature of the interaction but their magnitudes are constrained by the inequality in Equation (2). Geometrically, this inequality insures that our failure surface will intercept each stress axis. The shape of the surface will be ellipsoidal. The failure surface cannot be open-ended like a hyperboloid. Equation (2) makes sure that it will not happen. The same positive-definite requirement of F_{ij} is imposed on

F_i . The displacements of the ellipsoid due to internal stresses are such that the origin remains in the ellipsoid.

(7) Finally, Gol'denblat and Kopnov [4] was one of the first to suggest the use of strength tensors and proposed a general theory in the following form:

$$(F_i \sigma_i)^\alpha + (F_{ij} \sigma_i \sigma_j)^\beta + (F_{ijk} \sigma_i \sigma_j \sigma_k)^\gamma + \dots = 1 \quad (3a)$$

He investigated a special case of

$$\alpha = 1, \beta = 1/2, \gamma = -\infty \quad (3b)$$

The $\pm$ sign associated with the square root is awkward. This, however, can be eliminated by simple rearrangement of this special case and we will have:

$$2F_i \sigma_i + F_{ij} \sigma_i \sigma_j - (F_i \sigma_i)^2 = 1 \quad (4)$$

This relation is also more complicated than Equation (1). The additional term does not introduce any more generality than the linear and quadratic approximation of Equation (1); i.e., there are 6 linear and 21 quadratic terms. We believe that our approximation is operationally simpler and will be investigated in detail in this paper.

SYMMETRY PROPERTIES

The symmetry properties of our strength tensors follow well established patterns of the diffusion [5] and elastic [6] properties of anisotropic materials. For a triclinic material in 3-space:

$$F_i = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \\ F_6 \end{Bmatrix} \quad (5a)$$

$$F_{ij} = \begin{bmatrix} F_{11} & F_{12} & F_{13} & F_{14} & F_{15} & F_{16} \\ & F_{22} & F_{23} & F_{24} & F_{25} & F_{26} \\ & & F_{33} & F_{34} & F_{35} & F_{36} \\ & & & F_{44} & F_{45} & F_{46} \\ & & & & F_{55} & F_{56} \\ & & & & & F_{66} \end{bmatrix} \quad (5b)$$

We are assuming that both strength tensors are symmetric. The number of independent strength components are 6 and 21 for F_i and F_{ij} , respectively.

If a material has some form of symmetry, we would expect that a number of interaction terms will vanish. For specially orthotropic materials, for example, the off-diagonal terms in Equation (5) which are F_4 , F_5 and F_6 are

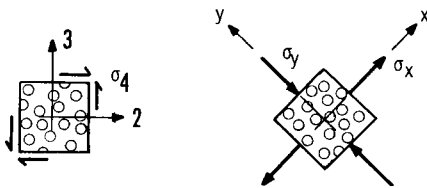


Figure 1. Equivalent states of stress in pure shear and tension-compression.

expected to vanish. The coupling between the normal and shear strengths, e.g., F_{16} , will also vanish if we assume that the change in the sign of shear stress does not change the failure stress. By essentially the same reasoning, we can assume that shear strengths for a specially orthotropic material are all uncoupled; i.e., $F_{45} = F_{56} = F_{64} = 0$. The coupling between normal

strengths, however, are expected to remain. With these assumed symmetry relations, the number of independent components reduced to 3 and 9, respectively.

For transversely isotropic material with plane 2-3 as the isotropic plane, we can immediately state that indices associated with this plane are identical;

$$F_2 = F_3, \quad F_{12} = F_{13}, \quad F_{22} = F_{33}, \quad \text{and} \quad F_{55} = F_{66} \quad (6)$$

We can also state that the two states of stress in Figure 1 are identical and should yield identical failure stresses, then

$$F_{44} = 2(F_{22} - F_{23}) \quad (7)$$

The number of independent components reduced to 2 and 5 for F_i and F_{ij} , respectively. The number of nonzero components for this transversely isotropic material remain as 3 and 9 as the case of special orthotropy.

By extending the relation of Equations (6) and (7) to the other two orthogonal planes, we obtain for isotropic materials 1 and 2 independent components for F_i and F_{ij} . If internal stresses or Bauschinger's effect is ignored, the only F_i component will vanish. If a failure by hydrostatic stresses is assumed to be inadmissible, the components of F_{ij} become related by

$$(F_{11} + F_{22} + F_{33}) + 2(F_{12} + F_{23} + F_{31}) = 0 \quad (8)$$

For isotropic materials, indices 1, 2 and 3 are identical, then Equation (8) becomes:

$$F_{12} = -F_{11}/2 \quad (9)$$

The three shear components are also identical and by virtue of Equation (9), we obtain

$$F_{44} = F_{55} = F_{66} = 2(F_{11} - F_{12}) = 3F_{11} \quad (10)$$

Thus, for isotropic materials which are plastically incompressible and zero internal stresses, the strength tensors are:

$$F_i \equiv \{0\} \quad (11a)$$

$$F_{ij} = F_{11} \begin{bmatrix} 1 & -1/2 & -1/2 & 0 & 0 & 0 \\ & 1 & -1/2 & 0 & 0 & 0 \\ & & 1 & 0 & 0 & 0 \\ & & & 3 & 0 & 0 \\ & & & & 3 & 0 \\ & & & & & 3 \end{bmatrix} \quad (11b)$$

For laminated composites, it may be easier to deal with a failure criterion in terms of strain components. Equation (1) can be rewritten as follows:

$$g(\epsilon_k) = G_i \epsilon_i - G_{ij} \epsilon_i \epsilon_j = 1 \quad (12)$$

where $G_i = F_m C_{mi}$
 $G_{ij} = F_{mn} C_{mi} C_{nj}$
 C_{ij} = Elastic Stiffness Matrix

When a state of plane stress is applied to the 1-2 plane, a triclinic material will appear as follows:

$$F_i = \begin{Bmatrix} F_1 \\ F_2 \\ F_6 \end{Bmatrix} \quad F_{ij} = \begin{bmatrix} F_{11} & F_{12} & F_{16} \\ & F_{22} & F_{26} \\ & & F_{66} \end{bmatrix} \quad (13)$$

There are a total of 9 independent strength components. For specially orthotropic materials,

$$F_6 = F_{16} = F_{26} = 0 \quad (14)$$

we have 2 and 4 independent components for F_i and F_{ij} , respectively.

ENGINEERING STRENGTHS

The relations between engineering strengths and strength tensors are similar to those between engineering constants and the components of the elastic compliance, typical relations for the latter are:

$$E_{11} = 1/S_{11}, \quad G_{12} = 1/S_{66} \quad (15)$$

where E_{11} is the longitudinal stiffness; G_{12} , the longitudinal shear modulus; S_{ij} , the components of the compliance matrix. An important point that has often been overlooked is that engineering constants are NOT components of a 4th rank tensor; while S_{ij} are components of such a tensor within a correction factor caused by use of the contracted notation.

Like engineering constants, engineering strengths are those strength parameters which are relatively simple to measure in the laboratory. Although they are not components of a tensor, they can be related to the components of F_i and F_{ij} through relations which we will establish and which

turn out to be similar to those in Equation (15). Let us, for example, impose a uniaxial tensile stress on a specimen oriented along the 1-axis. We can measure the tensile failure and designate the failure stress X . Similarly, we can experimentally obtain a uniaxial compressive strength along the same axis and call it X' . From these two simple experiments, we obtained two engineering strengths X and X' . They can be related to the strength components F_1 and F_{11} through Equation (1) if we let $i = 1$ only; i.e.,

$$F_{11} \sigma_1^2 + F_1 \sigma_1 = 1 \quad (16)$$

When $\sigma_1 = X$, Equation (16) becomes

$$X^2 F_{11} + X F_1 = 1 \quad (17)$$

When $\sigma_1 = -X'$, Equation (16) becomes

$$X'^2 F_{11} - X' F_1 = 1 \quad (18)$$

Solve Equations (17) and (18) simultaneously, we obtain

$$F_{11} = \frac{1}{X X'} \quad (19)$$

$$F_1 = \frac{1}{X} - \frac{1}{X'} \quad (20)$$

Through uniaxial tensile and compressive tests imposed along the 2- and 3-axis, we obtain

$$F_{22} = \frac{1}{Y Y'}, \quad F_2 = \frac{1}{Y} - \frac{1}{Y'} \quad (21)$$

$$F_{33} = \frac{1}{Z Z'}, \quad F_3 = \frac{1}{Z} - \frac{1}{Z'} \quad (22)$$

where Y and Y' are the uniaxial tensile and compressive strengths along the 2-axis; Z and Z' , those along the 3-axis.

By imposing pure shear in the 3 orthogonal planes we can obtain

$$F_{44} = \frac{1}{Q Q'}, \quad F_4 = \frac{1}{Q} - \frac{1}{Q'} \quad (23)$$

$$F_{55} = \frac{1}{R R'}, \quad F_5 = \frac{1}{R} - \frac{1}{R'} \quad (24)$$

$$F_{66} = \frac{1}{S S'}, \quad F_6 = \frac{1}{S} - \frac{1}{S'} \quad (25)$$

Where Q and Q' are positive and negative pure shear strengths along the 2-3

plane; R and R' , those along the 3-1 plane; and S and S' , those along the 1-2 plane.

We have thus far established all 6 components of F_i and all the diagonal components of F_{ij} . The off-diagonal components are related to the interaction of two stress components in the strength criterion. For example, the experimental determinations of F_{12} and F_{16} require combined stresses. Simple uniaxial or pure shear tests will not be sufficient. Most existing strength criteria do not require combined stress tests because the interactions term such as F_{12} is assumed to be a dependent quantity or F_{16} is zero. Strength component F_{12} can be determined by an infinite number of combined stresses; only a few simple combinations will be discussed here. If we impose a biaxial tension such as

$$\sigma_1 = \sigma_2 = P, \quad \sigma_3 = \sigma_4 = \sigma_5 = \sigma_6 = 0 \quad (26a)$$

Substitute this state of combined stresses into Equation (1), we obtain

$$P^2 (F_{11} + F_{22} + 2F_{12}) + P (F_1 + F_2) = 1 \quad (26b)$$

Solving for F_{12} , we obtain

$$F_{12} = \frac{1}{2P^2} \left[1 - P \left(\frac{1}{X} - \frac{1}{X'} + \frac{1}{Y} - \frac{1}{Y'} \right) - P^2 \left(\frac{1}{XX'} + \frac{1}{YY'} \right) \right] \quad (26c)$$

We can determine F_{23} and F_{31} by imposing biaxial tension or compression in the 2-3 and 3-1 planes, respectively.

Many Russian workers recommend the use of 45-degree specimens for the determination of interaction terms such as F_{12} . This can be done by letting:

$$\sigma_1 = \sigma_2 = \sigma_6 = U/2, \quad \sigma_3 = \sigma_4 = \sigma_5 = 0 \quad (27a)$$

where U is the tensile strength of a 45-degree off-axis specimen. Note that the combined stresses in Equation (27a) are applied to the symmetry axes of a specially orthotropic material. This state of stress is equivalent to a uniaxial tensile stress applied to a reference coordinate system rotated 45 degrees from the material symmetry axes. This is why U can be considered as an engineering strength. Care must be exercised in the actual experiment of loading a 45-degree specimen so that the shear coupling effect due to S_{16} is minimized. By introducing Equation (27a) into Equation (1), we have

$$\frac{U^2}{4} (F_{11} + F_{22} + 2F_{12} + F_{66}) + \frac{U}{2} (F_1 + F_2) = 1 \quad (27b)$$

where F_{16} , F_{26} , and F_6 vanish because of the assumed special orthotropy. Now

we can obtain F_{12} by this test that

$$F_{12} = \frac{2}{U^2} \left[1 - \frac{U}{2} \left(\frac{1}{X} - \frac{1}{X'} + \frac{1}{Y} - \frac{1}{Y'} \right) - \frac{U^2}{4} \left(\frac{1}{X X'} + \frac{1}{Y Y'} + \frac{1}{S^2} \right) \right] \quad (27c)$$

A similar relation for the compressive strength U' of a 45 degree off-axis unidirectional specimen is as follows:

$$\sigma_1 = \sigma_2 = \sigma_6 = -U'/2, \quad \sigma_3 = \sigma_4 = \sigma_5 = 0 \quad (28a)$$

$$\frac{U'^2}{4} (F_{11} + F_{22} + 2F_{12} + F_{66}) - \frac{U^2}{2} (F_1 + F_2) = 1 \quad (28b)$$

Then we can find

$$F_{12} = \frac{2}{U'^2} \left[1 + \frac{U'}{2} \left(\frac{1}{X} - \frac{1}{X'} + \frac{1}{Y} - \frac{1}{Y'} \right) - \frac{U'^2}{4} \left(\frac{1}{X X'} + \frac{1}{Y Y'} + \frac{1}{S^2} \right) \right] \quad (28c)$$

By comparing Equation (27b) and (28b), we can derive the following relations between U and U' :

$$2 \left(\frac{1}{U} - \frac{1}{U'} \right) = F_1 + F_2 = 2T_1 \quad (29a)$$

$$\frac{4}{U U'} = F_{11} + F_{22} + 2F_{12} + F_{66} = 4(U_1 - U_3) \quad (29b)$$

where T_1 is the first invariant of F_i which will be shown in Table I and Figure 3 of this paper; while the difference on the right-hand side of Equation (29b) is not invariant which can be seen from Table II.

Let V and V' be the positive and negative shear strengths of a 45-degree off-axis unidirectional specimen, then analogous to the relations in Equations (27) through (29) for uniaxial stresses, we have

$$+\sigma_1 = -\sigma_2 = V, \quad \sigma_3 = \sigma_4 = \sigma_5 = \sigma_6 = 0 \quad (30a)$$

This state of stress is applied to the symmetry axes oriented at +45 degrees from the positive shear stress V . The same state of stress exists when a negative shear stress ($-V$) is applied to a -45-degree off-axis specimen. Substituting Equation (30a) into (1), we have

$$V^2 (F_{11} + F_{22} - 2F_{12}) + V (F_1 - F_2) = 1 \quad (30b)$$

Then we can establish

$$F_{12} = -\frac{1}{2V^2} \left[1 - V \left(\frac{1}{X} - \frac{1}{X'} - \frac{1}{Y} + \frac{1}{Y'} \right) - V^2 \left(\frac{1}{XX'} + \frac{1}{YY'} \right) \right] \quad (30c)$$

Similarly, when

$$+ \sigma_1 = -\sigma_2 = -V', \quad \sigma_3 = \sigma_4 = \sigma_5 = \sigma_6 = 0 \quad (31a)$$

We have

$$V'^2 (F_{11} + F_{22} - 2F_{12}) - V' (F_1 - F_2) = 1 \quad (31b)$$

$$F_{12} = -\frac{1}{2V'^2} \left[1 + V' \left(\frac{1}{X} - \frac{1}{X'} - \frac{1}{Y} + \frac{1}{Y'} \right) - V'^2 \left(\frac{1}{XX'} + \frac{1}{YY'} \right) \right] \quad (31c)$$

From Equations (30b) and (31b) we can derive the following relations between V and V' :

$$\frac{1}{V} - \frac{1}{V'} = F_1 - F_2 = 2T_2 \quad (32a)$$

$$\frac{1}{VV'} = F_{11} + F_{22} - 2F_{12} \quad (32b)$$

All these relations are helpful in determining components of F_{ij} and their transformed quantities as well as the internal consistency of this present theory. From Equations (29b) and (32b), we can derive an invariant relation:

$$\frac{1}{VV'} - \frac{4}{UU'} = 4(U_5 - U_4) = F_{66} - 4F_{12} \quad (33)$$

We will describe later in this paper how component F_{12} for a given composite system, e.g., the graphite-epoxy composite, can be best determined. Suffice to say, F_{12} is a very sensitive and critical quantity in this proposed theory and must be understood by its users.

For anisotropic material, strength component F_{16} , which is no longer zero, can be determined by a tension-torque combination; e.g.,

$$\sigma_1 = \sigma_6 = T, \quad \sigma_2 = \sigma_3 = \sigma_4 = \sigma_5 = 0 \quad (34a)$$

This is equivalent to letting $i = 1$ and 6 in Equation (1), and we have

$$T^2 (F_{11} + F_{66} + 2F_{16}) + T (F_1 + F_6) = 1 \quad (34b)$$

by rearranging,

$$F_{16} = \frac{1}{2T^2} \left[1 - T \left(\frac{1}{X} - \frac{1}{X'} + \frac{1}{S} - \frac{1}{S'} \right) - T^2 \left(\frac{1}{XX'} + \frac{1}{SS'} \right) \right] \quad (34c)$$

This experiment can be readily performed by testing a tubular specimen with the tube axis along the 1-axis. Component F_{26} can be determined by imposing tension-torque on a tubular specimen with the 1-axis along the circumference of the tube. Similar to the case of biaxial tension where the ratio and signs of the two normal stresses may be arbitrary, the tension-torque combination can also have ratios and signs different from that of Equation (34a), where the ratio was unity and signs positive. Different ratios and/or signs will provide additional determinations of the interaction terms. The redundant measurements can be used to establish the range of validity and accuracy of our initial assumption stated in Equation (1). If the tube axis coincide with the material symmetry axis, e.g., the filament axis of a unidirectional composite runs along the longitudinal direction of the tube, F_{16} in Equation (34c) must be zero in this specially orthotropic orientation.

QUADRATIC APPROXIMATIONS

Using the framework and notation of our approach, we can compare the forms of many existing quadratic approximations of the strength criteria. The Hill criterion [2] which is limited to specially orthotropic 3-dimensional body, with plastic incompressibility and without internal stresses, can be expressed in the following forms:

$$F_i \equiv \{0\} \quad (35a)$$

$$F_{ij} = \begin{bmatrix} \frac{1}{X^2} & -\frac{1}{2} \left(\frac{1}{X^2} + \frac{1}{Y^2} - \frac{1}{Z^2} \right) & -\frac{1}{2} \left(\frac{1}{Z^2} + \frac{1}{X^2} - \frac{1}{Y^2} \right) & 0 & 0 & 0 \\ & \frac{1}{Y^2} & -\frac{1}{2} \left(\frac{1}{Y^2} + \frac{1}{Z^2} - \frac{1}{X^2} \right) & 0 & 0 & 0 \\ & & \frac{1}{Z^2} & 0 & 0 & 0 \\ & & & \frac{1}{Q^2} & 0 & 0 \\ & & & & \frac{1}{R^2} & 0 \\ & & & & & \frac{1}{S^2} \end{bmatrix} \quad (35b)$$

Since the Hill criterion is obtained by generalizing the Mises criterion, 3 interaction terms become dependent on the diagonal terms. Such arbitrary generalization brought in only 6 independent strength components in F_{ij} instead of 9. The Mises criterion can be recovered from Equation (35b) if we let

$$X^2 = Y^2 = Z^2 = Q^2/3 = R^2/3 = S^2/3 \quad (36)$$

Then Equation (35b) becomes the same as Equation (11b) with F_{11} replaced by $1/X^2$. Substituting the latter result into Equation (1), we have

$$\begin{aligned} &\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \sigma_1\sigma_2 - \sigma_2\sigma_3 - \sigma_3\sigma_1 \\ &+ 3(\sigma_4^2 + \sigma_5^2 + \sigma_6^2) = X^2 \end{aligned} \quad (37a)$$

or

$$\begin{aligned} &(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \\ &+ 6(\sigma_4^2 + \sigma_5^2 + \sigma_6^2) = 2X^2 = 6S^2 \end{aligned} \quad (37b)$$

Some authors have tried to generalize the quadratic approximation derived from Equation (35b) by introducing floating or adjustable constants for the off-diagonal or interaction terms such as [7]

$$F_{12} = K \left(\frac{1}{X^2} + \frac{1}{Y^2} - \frac{1}{Z^2} \right) \quad (38)$$

The use of constant "K" implies that F_{12} is proportional to a particular function of the engineering strengths which is in general not the case. As stated earlier, the generalization of the Mises criterion to describe special orthotropy lacks analytic foundation. Further generalization from the form of Equation (35b) by means of adjustable constants of proportionality may lead to unnecessarily restrictive, if not erroneous, strength criteria. The use of arbitrary constants, like K in Equation (38), does not insure internal consistency and invariance under transformation, and may violate the stability requirement of Equation (2).

In fact, when the strength criterion based on Equation (35b) is specialized to plane stress applied to a unidirectional composite with the longitudinal strength X in the 1-direction, the two transverse strengths are equal; i.e.,

$$Y = Z = \text{Transverse Strength} \quad (39)$$

With this additional assumption, we can derive the often-used strength criterion

$$\frac{\sigma_1^2}{X^2} - \frac{\sigma_1\sigma_2}{X^2} + \frac{\sigma_2^2}{Y^2} + \frac{\sigma_6^2}{S^2} = 1 \quad (40)$$

Note that this criterion is biased; i.e., the interaction term contains only the

longitudinal strength. If we are not careful and apply the plane-stress strength criterion to the same unidirectional composite but with filaments running in the 2-direction, we will obtain the following criterion:

$$\frac{\sigma_1^2}{Y^2} - \frac{\sigma_1\sigma_2}{Y^2} + \frac{\sigma_2^2}{X^2} + \frac{\sigma_6^2}{S^2} = 1 \quad (41)$$

This latter equation is not a correct specialization of the strength tensor in Equation (35b) because the equivalent relation of Equation (39) no longer holds. For the same reason, the plane-stress strength criterion cannot be applied to a laminate because the strength transverse to the plane of the laminate is an independent engineering strength, entirely unrelated to the in-plane engineering strengths.

TRANSFORMATION OF STRENGTH TENSORS

In order to gain insight into our strength criterion, it is helpful if we examine the transformed properties of the strength tensors. We will show graphically the transformed properties of unidirectional graphite-epoxy composites with the following engineering strengths:

$$\begin{aligned} X &= \text{Longitudinal Tension} = 150 \text{ ksi} \\ X' &= \text{Longitudinal Compression} = 100 \text{ ksi} \\ Y &= \text{Transverse Tension} = 6 \text{ ksi} \\ Y' &= \text{Transverse Compression} = 17 \text{ ksi} \\ S &= \text{Longitudinal Shear} = 10 \text{ ksi} \\ F_{12} &= \pm \sqrt{F_{11} F_{22}} = \pm .0008084 \end{aligned} \quad (42)$$

Assuming that the graphite unidirectional composite is specially orthotropic and under plane stress, the principal strength components are:

$$F_i = \begin{Bmatrix} F_1 \\ F_2 \\ F_6 \end{Bmatrix} = \begin{Bmatrix} -.003 \\ +.107 \\ 0 \end{Bmatrix} \quad (43)$$

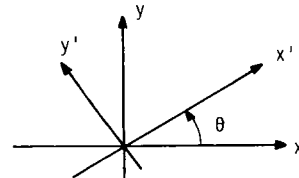


Figure 2. Direction of positive rotation of a coordinate transformation about the z- or 3-axis of a right-handed system.

$$F_{ij} = \begin{bmatrix} F_{11} & F_{12} & 0 \\ & F_{22} & 0 \\ & & F_{66} \end{bmatrix} = \begin{bmatrix} .00007 & \pm .00081 & 0 \\ & .0098 & 0 \\ & & .01 \end{bmatrix} \quad (44)$$

Where the range on F_{12} is imposed by the stability condition of Equation (2).

The transformation relations of F_i and F_{ij} in terms of multiple angles are shown in tabular form below, where angle θ is shown in Figure 2.

Table I. Transformation of F_i

	Constant	$\cos \theta$	$\sin \theta$	$\cos 2\theta$	$\sin 2\theta$
F_1'	T_1	0	0	T_2	F_6
F_2'	T_1	0	0	$-T_2$	F_6
F_6'	0	0	0	$2F_6$	$-2T_2$
F_4'	0	$2F_4$	$2F_5$	0	0
F_5'	0	$2F_4$	$2F_5$	0	0
F_3'	F_3	0	0	0	0

Where $T_1 = (F_1 + F_2)/2 = \text{Invariant}$

$$T_2 = (F_1 - F_2)/2$$

Table II. Transformation of F_{ij}

	Constant	$\cos 2\theta$	$\sin 2\theta$	$\cos 4\theta$	$\sin 4\theta$
F_{11}'	U_1	U_2	$2U_6$	U_3	U_7
F_{22}'	U_1	$-U_2$	$-2U_6$	U_3	U_7
F_{12}'	U_4	0	0	$-U_3$	$-U_7$
F_{66}'	$4U_5$	0	0	$-4U_3$	$-4U_7$
F_{16}'	0	$2U_6$	$-U_2$	$2U_7$	$-2U_3$
F_{26}'	0	$2U_6$	$-U_2$	$-2U_7$	$2U_3$
F_{13}'	V_1	V_2	$F_{36}/2$	0	0
F_{23}'	V_1	$-V_2$	$-F_{36}/2$	0	0
F_{36}'	0	F_{36}	$-V_2$	0	0
F_{44}'	W_1	W_2	$-F_{45}/2$	0	0
F_{55}'	W_1	$-W_2$	$F_{45}/2$	0	0
F_{45}'	0	$F_{45}/2$	W_2	0	0
F_{33}'	F_{33}	0	0	0	0

Where $U_1 = (3F_{11} + 3F_{22} + 2F_{12} + F_{66})/8 = \text{Invariant}$

$$U_2 = (F_{11} - F_{22})/2$$

$$U_3 = (F_{11} + F_{22} - 2F_{12} - F_{66})/8$$

$$U_4 = (F_{11} + F_{22} + 6F_{12} - F_{66})/8 = \text{Invariant}$$

$$U_5 = (F_{11} + F_{22} - 2F_{12} + F_{66})/8 = \text{Invariant}$$

$$U_6 = (F_{16} + F_{26})/4$$

$$U_7 = (F_{16} - F_{26})/4$$

$$V_1 = (F_{13} + F_{23})/2 = \text{Invariant}$$

$$V_2 = (F_{13} - F_{23})/2$$

$$W_1 = (F_{44} + F_{55})/2 = \text{Invariant}$$

$$W_2 = (F_{44} - F_{55})/2$$

The graphical representation of F_i' and F_{ij}' are shown in Figures 3 and 4, respectively. Components F_2' , F_{22}' and F_{26}' can be obtained from F_1' , F_{11}' and F_{16}' , respectively, by changing θ in Tables I and II to $-(\theta + 90)$, as shown in the Figures.

The transformed properties shown in Figures 3 and 4 are typical of 2nd and 4th rank tensors. The invariants associated with this particular transformation, i.e., a rotation about the 3-axis, are also shown as horizontal lines. The solid lines represent the upper bound of F_{12} , i.e., $F_{12} = +.0008$; and the dashed lines, the lower bound, i.e., $F_{12} = -.0008$. The present theory can only admit limited values of F_{12} for the stability reason depicted by Equations (2), (42) and (44).

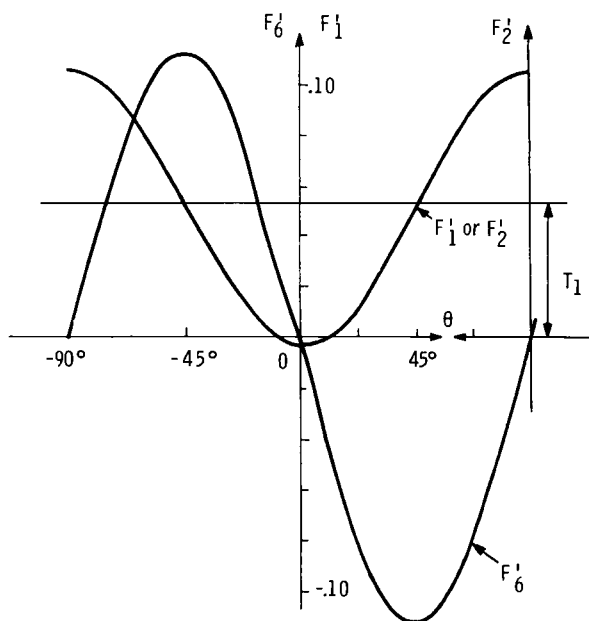


Figure 3. Transformation of strength tensor F_i for a graphite-epoxy composite system. Invariant T_1 represents the average value of the area under the F_1' and F_2' curves.

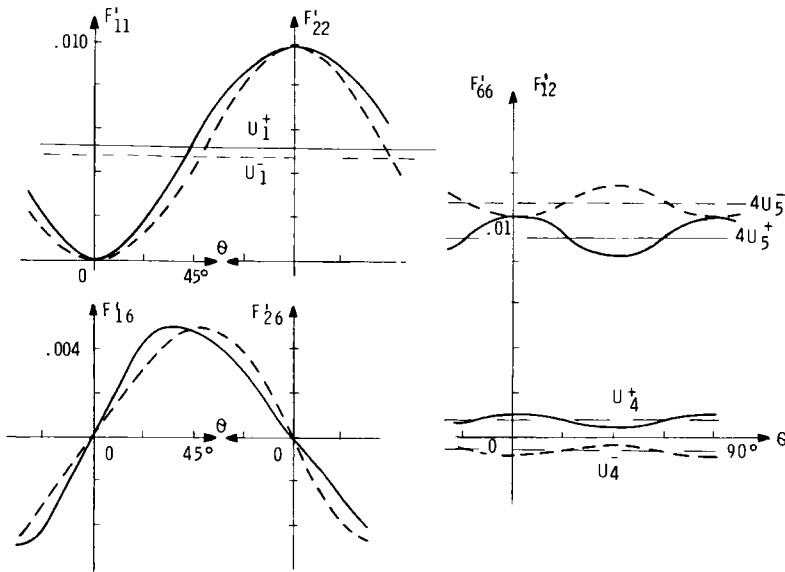


Figure 4. Transformation of F_{ij} for a graphite-epoxy composite. Solid lines represent the upper bound of F_{12} ; dashed lines, the lower bound. Invariants associated with various components are shown as horizontal lines. The plus and minus superscripts on U_i correspond to the bounds on F_{12} .

Similar constraint must be imposed on all the remaining interaction or off-diagonal terms of F_{ij} . If the magnitude of F_{12} , for example, exceeds the limits indicated by Equation (2), the resulting yield surface may become hyperboloidal. The off-axis properties in case of a F_{12} outside the stable range may have several maxima or minima which do not look like the expected well-behaved functions. In other cases, the off-axis properties "blow-up" at certain angles. This was shown by Ashkenazi [8] when he substituted an unconstrained value for F_{12} into the theory proposed by Gol'denblat and Kopnov [4]. Thus the initial assumption of Equation (1) in this paper and that employed in Reference 4 must be further constrained by some stability considerations. If experimental data do not agree with the predictions and constraints of our theory, we can modify the initial assumptions, such as a change in the functional form of Equation (1) or the inclusion of higher order terms, but we are not at liberty to relax the stability requirement.

The determination of the value of F_{12} can be achieved through infinite number of combined-stress states, shown earlier in this paper. The Russian workers [4, 8] suggested the use of off-axis tests, with the 45-degree specimen

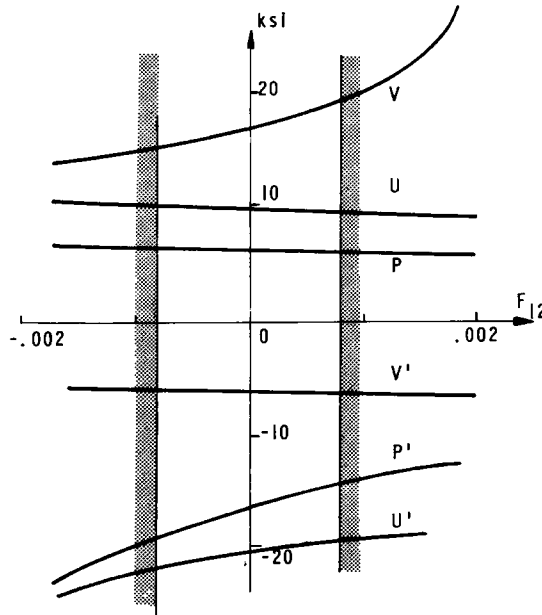


Figure 5. Effect of F_{12} on the combined-stress test data for graphite-epoxy composite. The bounds of F_{12} are shown. Quadratic approximations by Hill [2] and Hoffman [10] correspond to essentially zero value for F_{12} in this Figure.

as the most popular choice. It is interesting to observe the effect of these 45-degree uniaxial (U , U') and pure shear (V , V'), and hydrostatic (P , P') tests on the value of F_{12} . In Figure 5, we used the first five data for graphite-epoxy composites listed in Equation (42) and various values of F_{12} . These values are substituted into Equations (26) through (31) to obtain the curves in Figure 5 for various combined-stress properties U , U' , V , V' , P and P' . Curves such as U , P and V' are nearly horizontal which means that those tests should not be used for the determination of F_{12} . A small inaccuracy in the values of U , for example, can induce a large change in F_{12} . Of the remaining curves, the V (positive shear of a 45-degree specimen) gives the most reliable determination of F_{12} .

Also shown in Figure 5 are the limits imposed by the stability conditions of Equation (2). For the remaining part of this paper, we will use a F_{12} value close to its upper bound, say, $+.0008$, since the positive shear strength from our experimental measurement of a 45-degree specimen is about 20 ksi. It should be emphasized that Figure 5 is valid for a particular composite with engineering strengths shown in Equation (42). For other composites, we believe that similar study on the sensitivity of off-axis or other combined-stress tests on F_{12} should be made before a reasonable value of F_{12} is decided.

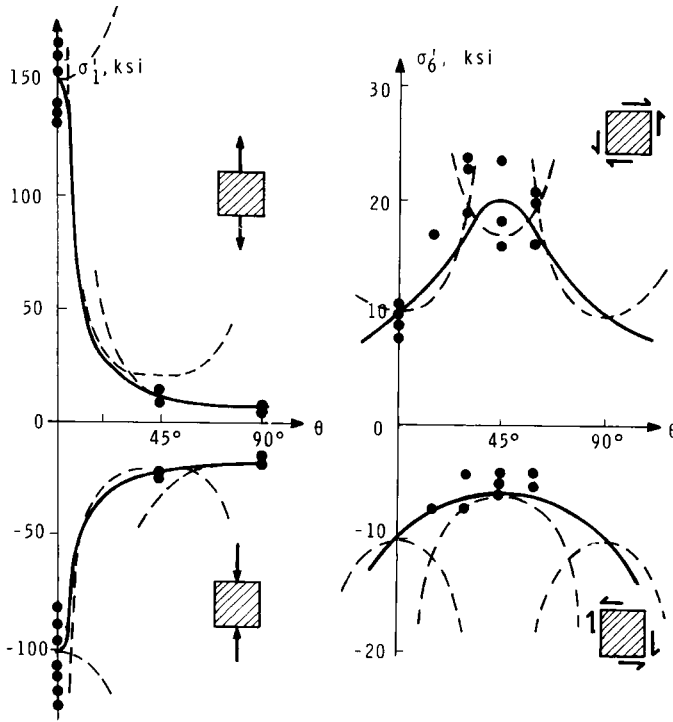


Figure 6. Off-axis uniaxial and shear strengths of graphite-epoxy composite. Solid lines represent our theory; dashed lines, the maximum stress theory; and dots, experimental data from tubular specimens [9].

OFF-AXIS PROPERTIES

If we want to obtain the off-axis uniaxial strengths, we simply use the transformed strength tensors in Equation (16); i.e.,

$$F_{11}' \sigma_1^2 + F_1' \sigma_1 - 1 = 0 \quad (45)$$

The transformed strength tensors are shown in Figures 3 and 4. At 45 degrees, for $F_{12} = +.0008$ for example,

$$\begin{aligned} F_{11}' &= .00537 \\ F_1' &= .05225 \end{aligned} \quad (46)$$

Solving the quadratic equation, we obtain two roots:

$$\sigma_1 = +9.61 \text{ and } -19.35 \text{ ksi} \quad (47)$$

If the process is repeated for other angles between zero and 90 degrees, we obtain the off-axis uniaxial shown in Figure 6. Like the transformed Young's modulus E_{11}' , the off-axis uniaxial properties do not follow the transformation of any known tensors. They are therefore by definition not tensors.

As another comparison, we also show in dashed lines in Figure 6 the prediction of the maximum stress theory which can be expressed analytically by one of the following 6 criteria whichever happens to be minimum for a given state of plane stress:

For uniaxial tensions:

$$\sigma_1 = X/m^2, Y/n^2, \text{ or } S/mn \quad (48)$$

For uniaxial compression:

$$\sigma_1 = X'/m^2, Y'/n^2, \text{ or } S'/mn \quad (49)$$

where $m = \cos \theta$, $n = \sin \theta$. The values of the engineering strengths are the same as those shown in Equation (42), except that only 5 of the 6 strengths are needed for the maximum stress theory. The biaxial tension P , for example, is assumed to be derivable from the 6 strength criteria shown in Equations (48) and (49). The predictions based on these criteria are shown as dashed lines in Figure 6.

The off-axis pure shear properties can also be obtained by solving the following quadratic equation for various values of angle θ :

$$F_{66}' \sigma_6^2 + F_6' \sigma_6 - 1 = 0 \quad (50)$$

Where the transformed strength tensors can be obtained from Figures 3 and 4. The results for the graphite composite with the same principal strengths in Equation (42) are shown in Figure 6. The corresponding predictions based on the maximum stress theory are shown as dashed lines and they are derived from the following relations:

For positive shear:

$$\sigma_6 = X/2mn, Y'/2mn, \text{ or } S/(m^2 - n^2) \quad (51)$$

For negative shear:

$$\sigma_6 = X'/2mn, Y/2mn, \text{ or } S'/(m^2 - n^2) \quad (52)$$

Although we have not shown the strength prediction based on the maximum strain theory in Figure 6, it is very similar in nature to the prediction of the maximum stress theory. There will be 6 simultaneous criteria for the case of plane stress. The lowest predicted strength among the 6 relations will govern for each state of combined stresses. The final criterion will consist of segmented curves as those shown in Figure 6. Each stress or strain criterion is unaffected by the presence or absence of other stress or strain components; i.e., there is no interaction among the stress or strain components.

From the shapes of the off-axis uniaxial and shear curves in Figure 6, we should be able to fit available test data very closely. Data points measured directly from graphite-epoxy tubular specimens by Halpin and Wu [9] are

shown in the Figure. Without the analytic foundation contained in the initial postulate in Equation (1), it is very difficult, if not impossible, to deduce the transformation properties associated with the strength of a composite from experimental data like those shown in Figure 6. The strength criterion proposed here certainly contains greater flexibility than most existing criteria while maintains the necessary generality in spacial dimensions, materials symmetries, combined-stress state, invariance, and, above all, operational simplicity.

TRANSVERSE SHEARS

As applications of composites for primary structures increase, the cross-sectional thickness is also increasing. The effects of transverse shears σ_4 and σ_5 in a thick laminate have been the subject of several recent articles, for example, by N. J. Pagano and others which have appeared in this Journal. If we confine ourselves to unidirectional layers oriented at various angles through a rotation about the 3-axis from the specially transversely isotropic orientation, our strength tensors in terms of engineering strengths are as follows:

$$F_i = \left\{ \begin{array}{c} \frac{1}{X} - \frac{1}{X'} \\ \frac{1}{Y} - \frac{1}{Y'} \\ \frac{1}{Y} - \frac{1}{Y'} \\ 0 \\ 0 \\ 0 \end{array} \right\} \quad (53a)$$

$$F_{ij} = \left[\begin{array}{cccccc} \frac{1}{XX'} & F_{12} & F_{12} & 0 & 0 & 0 \\ & \frac{1}{YY'} & F_{23} & 0 & 0 & 0 \\ & & \frac{1}{YY'} & 0 & 0 & 0 \\ & & & 2\left(\frac{1}{YY'} - F_{23}\right) & 0 & 0 \\ & & & & \frac{1}{S^2} & 0 \\ & & & & & \frac{1}{S^2} \end{array} \right] \quad (53b)$$

We have assumed in the above that the 2-3 plane is the isotropic plane. Only the interaction terms F_{12} and F_{23} are unfamiliar quantities. There is however no intrinsic difficulty associated with their experimental determination.

Application of our strength criterion to a 3-dimensional transversely isotropic material can be achieved in two ways. First, we can transform the stress components to the material-symmetry axes, in which case Equation (1) becomes

$$F_1 \sigma_1' + F_2 (\sigma_2' + \sigma_3') + F_{11} \sigma_1'^2 + F_{22} (\sigma_2'^2 + \sigma_3'^2 + 2\sigma_4'^2) + F_{66} (\sigma_5'^2 + \sigma_6'^2) \\ + 2F_{12} (\sigma_1' \sigma_2' + \sigma_1' \sigma_3') + 2F_{23} (\sigma_2' \sigma_3' - \sigma_4'^2) = 1 \quad (54)$$

Conceptually, this strength criterion is no more difficult than that for the plane stress. The most difficult job is the analysis that determines the additional stress components σ_3 , σ_4 , and σ_5 .

The other method of applying the strength criterion is to leave the stress components unchanged but rotate the material symmetry axes. Since we will have more strength components in the generally transversely isotropic configuration than the 7 independent components in Equation (54), Equation (1) when expanded for this case will have more terms. There will be the following 17 terms to be exact:

$$F_1' \sigma_1 + F_2' \sigma_2 + F_3' \sigma_3 + F_6' \sigma_6 + F_{11}' \sigma_1^2 + F_{22}' \sigma_2^2 + F_{33}' \sigma_3^2 \\ + F_{44}' \sigma_4^2 + F_{55}' \sigma_5^2 + F_{66}' \sigma_6^2 + 2F_{12}' \sigma_1 \sigma_2 + 2F_{23}' \sigma_2 \sigma_3 \\ + 2F_{31}' \sigma_3 \sigma_1 + 2F_{16}' \sigma_1 \sigma_6 + 2F_{26}' \sigma_2 \sigma_6 + 2F_{36}' \sigma_3 \sigma_6 \\ + 2F_{45}' \sigma_4 \sigma_5 = 1 \quad (55)$$

Again, this strength criterion is conceptually simple. The approach outlined in this section can be applied to 3-dimensionally-reinforced composites in general.

CONCLUSIONS

An operationally simple strength criterion can be developed using a scalar function of two strength tensors. This criterion is an improvement over most existing quadratic approximations of the yield surface because of the use of strength tensors. We can extend our working experience of elastic moduli and compliance to the strength criterion. The transformation relations and the associated invariants are well known. The number of independent and non-zero strength components for each material symmetry are also simple extensions of those governing the elastic properties of anisotropic materials. Spatial dimensions of the body and the states of combined stresses can be treated in a unified manner. Although available experimental data in com-

bined-stress states are only tentative, our strength criterion should fit them better than most, if not all, existing curve-fitting schemes. We are in the process of generating more data and it is our hope that the additional data will demonstrate more conclusively the utility of our criterion than we are able to do at this time.

Finally, we have tried to show the interaction term F_{12} as an independent but constrained strength component. It is an essential variable that makes the strength tensor possible. For the graphite-epoxy composite presented in this paper, the quadratic approximation of the strength criterion proposed by Hoffman [10], which is an improvement over Hill's [2] by incorporating the effect of internal stresses, has an interaction term so small ($-.00007$) that it can be treated as zero. This can be seen in Figure 5. For highly anisotropic composites, the off-axis tensile test does not appear to be an effective way to compare the difference between various strength theories, which include the maximum stress, Hoffman and ours. Off-axis compression data would be better. A positive shear such as V in Figure 5 is the most discriminating test.

ACKNOWLEDGMENT

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NOMENCLATURE

F_i	Strength tensor of the 2nd rank
F_{ij}	Strength tensor of the 4th rank
m, n	$\cos \theta, \sin \theta$
P, P'	Plane hydrostatic tensile and compressive stresses
Q, Q'	Positive and negative shear strengths in 2-3 plane
R, R'	Positive and negative shear strengths in 3-1 plane
S, S'	Positive and negative shear strengths in 1-2 plane
U, U'	45-degree tensile and compressive strengths in 1-2 plane
V, V'	45-degree positive and negative shear strengths in 1-2 plane
X, X'	Tensile and compressive strengths along the 1-axis
Y, Y'	Tensile and compressive strengths along the 2-axis
Z, Z'	Tensile and compressive strengths along the 3-axis
T_i	Transformation equations for F_i (Table I)
U_i, V_i, W_i	Transformations for equations for F_{ij} (Table II)
σ_i	Stress components
ϵ_i	Strain components

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